Few-Shot LLMs versus Domain-Adapted Encoders for Named Entity Recognition in Decisions from a Brazilian Court of Accounts

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Abstract. Decisions issued by Brazilian Courts of Accounts impose fines, obligations, reimbursements, and recommendations that staff transcribe into monitoring registries. We present `decicontas.br`, a public Named Entity Recognition (NER) dataset of 861 decisions of the Court of Accounts of the State of Rio Grande do Norte (TCE/RN) with four entity types—`MULTA` (fine), `OBRIGACAO` (obligation), `RESSARCIMENTO` (reimbursement), and `RECOMENDACAO` (recommendation)—and a comparative evaluation of nineteen extraction models. The dataset includes a confident-learning audit with paired releases and a three-annotator agreement study over the whole corpus (Fleiss $\kappa = 0.879$; pairwise span F1 0.824–0.838). Nine few-shot LLMs are compared with ten supervised baselines, including seven legal- or government-domain encoders, all reading complete documents, under a shared span-scoring protocol (IoU ≥ 0.5) with paired bootstrap inference and Holm correction over all 171 model pairs. DeepSeek-V4-Flash achieves macro span F1 0.799, exceeding the strongest supervised baseline, LegalBert-pt (0.613), by $+0.185$ (95% CI $[+0.097, +0.261]$, $p_{\text{Holm}} < 0.001$); its difference from GPT-4.1 (0.749) does not survive the correction. Rescored against each of the three annotations, the same predictions keep the leader, with gaps of $+0.185$ to $+0.265$ over the best supervised model. The ranking depends on the matching criterion: under exact span matching the encoders lead (0.522 against 0.373), so the LLM advantage holds for locating commands for human review, not for automatic delimitation. Data and code: github.com/eduardoplima/decicontas.br.

Keywords. Annotated Legal Corpus, Named Entity Recognition, Large Language Models, Brazilian Portuguese, Legal Informatics, Courts of Accounts, Few-Shot Learning, Inter-Annotator Agreement, Label-Error Detection

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1. Introduction

1.1. Unstructured Decisions and Accountability Gaps in Courts of Accounts

Named Entity Recognition (NER) underpins document analysis and compliance monitoring in the legal domain, but faces specialized terminology, long sentences, and entity types absent from general-purpose inventories [1], which has motivated a family of legal corpora and domain-adapted models for Brazilian Portuguese [1,2,3]. Brazilian Courts of Accounts (*Tribunais de Contas*) exercise external control over public administration. Their collegiate decisions impose pecuniary sanctions, enforceable obligations, reimbursement orders, and non-binding recommendations, under a constitutional mandate that gives debt and fine rulings the force of enforceable titles (Art. 71, § 3 of the 1988 Federal Constitution [4]). These directives, however, are published as free text, and over 93% of Brazilian Courts of Accounts disseminate their determinations in fragmented and formalistic ways, hampering follow-up and accountability [5]. At the Court of Accounts of the State of Rio Grande do Norte (TCE/RN), the institution studied here, internal rules mandate a General Registry for Decision Monitoring (CGAD), whose subregistries of fines, refunds, and recommendations are still fed by clerks who read each decision and manually transcribe the relevant passages. In a 2024 survey, 60% of these courts reported AI-based solutions in use and 53% listed automating repetitive tasks as a strategic goal [6]. Manual transcription is error-prone, and every error propagates to the registry that grounds enforcement follow-up.

Related work on Courts of Accounts includes hybrid classification of Italian *Corte dei Conti* decisions [7] and information retrieval over Brazilian federal audit-court decisions [8]. Our focus is the span-level extraction of commands for institutional monitoring.

1.2. Objectives and Contributions

This paper evaluates extraction configurations for identifying and typing the decision elements that feed the CGAD. We contribute **a public dataset with an explicit quality protocol**: 861 decisions annotated with four audit entities, a confident-learning audit [9] with pre- and post-review releases, and independent re-annotation of the whole corpus by two domain experts. We provide **a comparative evaluation of nineteen models**: nine few-shot LLMs and ten supervised baselines, including seven legal- or government-domain encoders, all reading complete documents and scored under a shared span-level protocol with paired document-level bootstrap and Holm correction [10]. We also provide **a sensitivity analysis to reference annotation**, rescoring the same predictions against all three annotations, alongside exact-match, model-selection, and prompting analyses and an error analysis linked to institutional review.

2. Related Work

2.1. NER Corpora for Brazilian Legal Text

LeNER-Br [1] was the first public NER corpus for Brazilian legal text; UlyssesNER-Br [2] shifted to the legislative domain; and CDJUR-BR [3] advanced annotation rigor over

a state court's procedural documents, with expert double annotation (Cohen's $\kappa = 0.69$) and adjudication. Closest to our setting, Menezes-Neto et al. introduced LexCare.br and evaluated few-shot LLMs for healthcare-related NER in Brazilian judicial decisions [11], whose prompting-based evaluation we extend with supervised baselines and statistical inference. Within the AI&Law community, Angelidis et al. [12] built an NER corpus and pipeline for Greek legislation, and Costa et al. [13] proposed a semantic annotation pipeline for Brazilian judicial decisions—the closest precedent in venue and material, though targeting ontology-driven annotation rather than a released NER benchmark. Beyond NER, LegalBench-BR [14] contributes a benchmark of Brazilian legal decision classification tasks for LLM evaluation. JurisTCU [8], 16,045 decisions of the federal Court of Accounts, is the closest audit-court resource, but it was built for information retrieval and carries no span annotation of the commands a decision issues.

2.2. Portuguese Legal and Governmental Encoders

BERTimbau [15] remains the reference Portuguese encoder, with a family of domain-adapted variants around it: JurisBERT and BERTimbauLaw on Brazilian legislation and case law [16], LegalBert-pt on documents from ten state courts [17], Legal-BERTimbau on European Portuguese supreme-court text [18], GovBERT-BR on governmental text [19], and RoBERTaLexPT [20].

2.3. LLMs for Named Entity Recognition

Few-shot prompting allows LLMs to extract from a handful of in-context demonstrations, without weight updates [21]. GPT-NER recasts NER as generation with markup tokens [22], and chain-of-thought prompting [23] adds intermediate reasoning, with mixed evidence in extraction tasks. In the legal domain, Lee et al. [24] applied few-shot learning to legal argument NER, and Adhikary et al. [25] used few-shot LLMs to extract case attributes for downstream prediction. Throughout we use *function calling*, in which the model fills the typed arguments of a client-declared function. Comparative studies in specialized domains report LLMs approaching or surpassing fine-tuned encoders in low-annotation regimes [11], but rarely control the prompting technique, quantify uncertainty, or test whether their conclusions survive a change of reference annotation.

3. The Decicontas.br Dataset

3.1. Source and Target Entities

The dataset comprises 861 decisions from collegiate sessions of the TCE/RN (Section 1.1), drawn from the institutional database. Four entity types mirror the CGAD subregistries [26].

`MULTA` (fine) is a pecuniary sanction for administrative or fiscal misconduct; `OBRIGACAO` (obligation) a binding duty imposed on the audited entity, including mandatory determinations; `RESSARCIMENTO` (reimbursement) a mandated reimbursement to the public treasury for established losses; and `RECOMENDACAO` (recommendation) a formally documented, non-binding recommendation addressed to an audited entity to improve its administrative controls (the guideline excludes recommendations addressed

to units of the Court itself). These are legal institutes with distinct enforcement regimes, not arbitrary NLP labels: fines and reimbursement rulings constitute enforceable titles [26], obligations bind under penalty of sanction, and recommendations feed future audit planning. Extraction errors therefore carry asymmetric legal costs—a missed fine may leave a sanction unregistered, while a spurious extraction may create an unfounded registry entry—which motivates both the strict annotation guidelines and the mandatory human review in deployment.

The corpus comprises 861 documents: 202 contain at least one entity, while the remaining 659 (76.5%) are true negatives — filings and clean-account judgments with no registrable command. All were read and validated; keeping them reflects the class prior of the production task, where a system must also learn *not* to extract. The corpus carries 411 entities: 203 `MULTA`, 123 `OBRIGACAO`, 63 `RESSARCIMENTO` and 22 `RECOMENDACAO`. Spans are long (per-class median lengths between 248 and 357 characters), which motivates the span-level evaluation with partial matching adopted in Section 4.2.

Documents are short and formulaic: 135.7 whitespace tokens on average, rising to 242.9 in the informative subset. Each class is anchored by strong lexical markers: the stem *ressarc-* appears within the first five tokens of 60 of the 63 `RESSARCIMENTO` spans (95.2%) and *recomend-* within the first five of 20 of the 22 `RECOMENDACAO` spans (90.9%), while the corresponding figures for `MULTA` (*mult-*, 166/203) and `OBRIGACAO` (*determin-*, 82/123) are markedly lower at 81.8% and 66.7%. These are frequent but individually insufficient cues: the same vocabulary occurs in non-dispositive passages that must *not* be extracted, which is precisely where models overextract (Section 5.6).

3.2. Annotation and Guidelines

The reference annotation was produced by the first author, an External Control Auditor at the Rio Grande do Norte State Court of Accounts, in Label Studio under deliberately strict guidelines: spans must be literal copies of the source and inference is forbidden, because extracted passages are later re-located by string matching and feed official registries that admit no interpolated content. Tokenization is canonical whitespace splitting, shared by every producer and consumer of the released files.

3.3. Label-Error Audit with Confident Learning

Annotation errors were audited with confident learning [9], which estimates the joint distribution of observed and latent true labels from out-of-sample predicted probabilities and flags the examples whose observed label is most likely wrong. The probabilities came from an ensemble of two complementary token models (a linear classifier over Portuguese BERT embeddings and a fine-tuned Portuguese BERT token classifier). The audit flagged 794 suspicious token groups; the 567 groups with ensemble confidence ≥ 0.95 were manually reviewed one by one, and the remaining 227 kept their original labels. Among the 567 reviewed groups, 23 (4.1%) were changed and the remaining suggestions were rejected. This proportion describes the reviewed subset, not an estimated error rate for the entire corpus. Its flagging profile is dominated by suspected omissions—of the 4,650 flagged tokens, the largest blocks are `O` $\rightarrow$ `OBRIGACAO` (640) and `O` $\rightarrow$ `RECOMENDACAO` (547)—but most of these suggestions did not survive human review. Both the pre- and post-review versions are published as paired releases (JSON,

Table 1. Inter-annotator agreement over the 861 documents. A1 is the post-audit gold; A2/A3 are independent expert annotators.

Pair	Token κ	Span F1 (micro)	Span F1 (macro)
A1 × A2	0.864	0.824	0.795
A1 × A3	0.875	0.828	0.792
A2 × A3	0.899	0.838	0.812
Fleiss κ (3-way)	0.879	—	—

JSONL, CoNLL, and BRAT formats with checksums), so that any result can be reproduced against either gold standard; all experiments in this paper use the post-review version (release 1.2.0), in which the names of the natural persons who are parties to the proceedings are pseudonymised, one token per token. The corpus is maintained alongside the Court's human-in-the-loop registry pipeline and undergoes periodic revisions and corrections, which may change entity counts between releases; each release is recorded in the repository.

3.4. Inter-Annotator Agreement

To quantify the reliability of the schema and guidelines, the full corpus was independently re-annotated under a blind protocol by two staff members of the Court's unit responsible for feeding the CGAD—domain experts in exactly the task the schema formalizes—with the post-review author annotation as the third version. Agreement is computed over all 861 documents at three granularities: token-level Cohen's κ per pair and Fleiss' κ overall [27], span-level pairwise F1 under the same matcher and the same token units as the model evaluation, and document-level presence. Token-level κ ranges from 0.864 to 0.899 with Fleiss $\kappa = 0.879$—"almost perfect" on the Landis–Koch scale [28]—and pairwise span F1 from 0.824 to 0.838 (Table 1), providing an empirical human-agreement reference. Class confusions are rare (13 cases across the three pairs, ten of them between the two pecuniary institutes `MULTA` and `RESSARCIMENTO`). Boundary divergences often reflect granularity: in document 95, for instance, the ruling imposes a single sanction of R$ 10,000.00 and then itemizes it into two R$ 5,000.00 parcels; the gold follows the ruling and annotates one `MULTA` span, whereas both independent annotators annotated one span per parcel. The dominant divergence is *registration*: for `RECOMENDACAO`, the two independent annotators recorded more recommendations than the gold (33 and 28 spans against 22; three of Annotator 3's address the Court's own units and fall outside the guideline), giving pairwise F1 of 0.764 and 0.760 against it but 0.820 between themselves, and document-presence κ of 0.704 and 0.693 against it but 0.778 between themselves. No matched span is confused between `OBRIGACAO` and `RECOMENDACAO`, suggesting that registrability decisions are a major source of disagreement. A robustness check against the pre-review gold (κ 0.853/0.866) also shows substantial token-level agreement.

4. Experimental Setup

4.1. Models

LLMs (few-shot). Nine instruction-tuned models: `gpt-4.1`, `gpt-4.1-mini`, `gpt-4.1-nano`, `gpt-5-mini`, `gpt-5.1` and `gpt-5.2` (OpenAI), the open-weights `deepseek-v4-flash` [29] (MIT), `llama-3.3-70b` (Meta) and `qwen2.5-72b` (Alibaba). All were consumed through managed APIs rather than self-hosted: the OpenAI models and DeepSeek via Azure AI Foundry (Brazil South), the other two via OpenRouter. Dated snapshots were pinned where exposed. Decoding used temperature 0, top-p 1, a 4096-token output limit and seed 1007 on Azure, where a seed is honoured on a best-effort basis and does not guarantee bit-identical output; OpenRouter exposes no seed, and the three `gpt-5.x` are reasoning models whose API ignores sampling parameters. Structured output was obtained via function calling against a typed schema mirroring the registry: one list per entity class, each item carrying a description field whose value must be a literal excerpt of the decision. The prompt embeds twelve curated exemplars, four with empty expected output, teaching the model not to extract from non-registrable commands. The prompt was frozen and shared across models in the main comparison. Beyond static few-shot, two techniques are evaluated on five representative LLMs: *chain-of-thought* [23] and a *two-stage* variant that first lists candidate passages and then structures them in a second call.

Supervised baselines. Ten models trained with 5-fold cross-validation: a BiLSTM-CRF [30], the generic BERTimbau base and large [15], and seven domain-adapted encoders [18,16,17,31,32,19]. Encoders read each document in windows of 512 subwords overlapping by 320; a word covered by two windows takes the prediction of the window in which it lies farthest from an edge. The BiLSTM-CRF accepts 1,100 tokens, above the longest document (1,056), so no model truncates its input. Out-of-fold predictions feed the comparison on the same 861 documents. For each model, the configuration (encoders: learning rate $\{2,3,5\} \times 10^{-5}$ and warmup ratio $\{0,0.1\}$; BiLSTM-CRF: eight combinations of hidden size, dropout and learning rate) was selected by span micro F1 on a single 80/20 development split, and checkpoints by strict IOB2 F1.

4.2. Metrics and Statistical Protocol

We report token-level F1 over collapsed labels and span-level precision, recall, and F1 in micro and macro aggregations. Adapting entity-level partial-matching evaluation [33], a prediction matches a gold entity of the same class when their IoU is ≥ 0.5; greedy bipartite matching uses each prediction and each entity at most once. **All predictions are scored in one unit**: spans are projected onto canonical whitespace token indices before matching, and a token belongs to a span whenever the two overlap, using the same rule as the gold BIO conversion. Every model is scored against the full gold of every document. A deterministic filter, applied identically to every model's predictions before scoring, discards predictions on deliberations addressed to units of the Court itself; no reference entity overlaps these clauses, and no discarded prediction overlapped one.

LLM outputs are strings, so they are first re-located in the decision. A string that occurs literally is placed at that occurrence (the one nearest the fuzzy anchor when it repeats); only non-literal strings fall back to fuzzy search, a 500-character window slid-

Table 2. Overall results on `decicontas.br` (861 documents, 411 entities). Span metrics use IoU ≥ 0.5 over whitespace token indices; P and R are span precision and recall. Sorted by macro span F1. †Leader's group: not distinguishable from DeepSeek-V4-Flash after Holm correction over the 171 pairs.

Model	Type	Token F1	P	R	$F1_\mu$	$F1_M$
DeepSeek-V4-Flash†	LLM	**0.869**	0.813	0.837	**0.825**	**0.799**
GPT-4.1†	LLM	0.841	0.700	0.859	0.772	0.749
GPT-5.1†	LLM	0.816	0.672	0.854	0.752	0.748
GPT-4.1-mini	LLM	0.825	0.638	0.881	0.740	0.719
GPT-5.2	LLM	0.790	0.555	0.871	0.678	0.692
LegalBert-pt	Sup.	0.796	0.823	0.691	0.751	0.613
Qwen2.5-72B	LLM	0.749	0.586	0.788	0.672	0.605
BERTimbau-large	Sup.	0.797	0.785	0.691	0.735	0.605
GPT-5-mini	LLM	0.601	0.299	**0.912**	0.450	0.583
BiLSTM-CRF	Sup.	0.813	0.825	0.676	0.743	0.567
Legal-BERT-STF	Sup.	0.775	0.811	0.659	0.728	0.552
GovBERT-BR	Sup.	0.748	0.795	0.642	0.711	0.539
JurisBERT	Sup.	0.747	0.797	0.640	0.710	0.532
Legal-BERTimbau-base	Sup.	0.750	0.807	0.633	0.709	0.530
BERTimbauLaw	Sup.	0.761	0.838	0.640	0.726	0.527
BERTimbau-base	Sup.	0.750	**0.853**	0.635	0.728	0.524
LegalBERTPT-br	Sup.	0.734	0.790	0.613	0.690	0.518
GPT-4.1-nano	LLM	0.623	0.388	0.630	0.481	0.435
Llama-3.3-70B	LLM	0.416	0.642	0.248	0.358	0.250

ing by 100 characters and scored by normalized Levenshtein partial ratio (≥ 80). For the leader, 395 of 479 strings (82.5%) are literal copies and none fails to align; failures are zero for six of the nine LLMs and peak at 1.4% for GPT-4.1-nano. Strings that cannot be located are discarded before scoring, so the metrics describe validated output.

Macro span F1 is the primary metric: it assigns equal weight to the four entity classes. Uncertainty is quantified by paired document-level bootstrap (10,000 resamples, 95% percentile CIs, two-sided p-values); reported differences are observed differences computed before rounding. Holm correction [10] is applied over all 171 model pairs. A non-significant difference is not evidence of equivalence.

Two asymmetries remain. First, the supervised configuration was selected once per model on the global 80/20 split, *not* within each cross-validation training fold, so the 173 development documents reappear in evaluation (20.1% of each test fold on average); Section 5.3 rescores every model without them. Second, exposure to negatives differs: about three quarters of the documents the encoders train on are negative (76.5% of the corpus), whereas only four of the twelve prompt exemplars are; the ratio in the prompt was not varied.

5. Results and Discussion

5.1. Overall Performance

Table 2 presents the nineteen models. Three findings organize the picture.

The strongest few-shot LLM has the highest macro span F1. DeepSeek-V4-Flash (0.799) exceeds the strongest supervised model, LegalBert-pt (0.613), by +0.185 (95% CI $[+0.097, +0.261]$; $p < 0.001$, $p_{\text{Holm}} < 0.001$), and the generic BERTimbau-large (0.605) and the BiLSTM-CRF (0.567) by +0.194 and +0.231 (both $p_{\text{Holm}} < 0.001$).

The gap is larger for macro than for micro averaging (+0.074; LegalBert-pt is fourth overall on micro F1) and is concentrated on RECOMENDACAO, as detailed below. A single training-set size does not establish the cause or generality of this benefit.

An open-weights model leads, without a significant difference from GPT-4.1 or GPT-5.1. DeepSeek-V4-Flash exceeds GPT-4.1 (0.749) by +0.049 (95% CI $[+0.016, +0.083]$, $p = 0.002$), a difference that does not survive Holm correction ($p_{\text{Holm}} = 0.192$); GPT-5.1 (0.748) likewise remains in the leader's group. Of the 171 model pairs, 115 differ at the nominal 5% level and 84 after Holm correction.

The top supervised models are not significantly different from one another. LegalBert-pt (0.613), BERTimbau-large (0.605) and the BiLSTM-CRF (0.567) lead the supervised group, and LegalBert-pt's differences from BERTimbau-large (+0.009, $[-0.047, +0.056]$) and the BiLSTM-CRF (+0.046, $[-0.008, +0.099]$) have intervals containing zero; none of the 45 supervised pairs survives Holm correction. The generic BERTimbau-base (0.524) is ninth of ten, within the band of the remaining domain-adapted encoders (0.518–0.552). This ranking does not isolate the effect of domain pre-training. No fold yields F1 = 0, and per-fold standard deviations of span F1 range from 0.048 to 0.133. Supervised models favor precision (0.785–0.853) over recall (0.613–0.691), whereas the competitive LLMs favor recall. This profile motivates evaluating LLMs as pre-annotators, although the relative effort of correcting insertions and omissions was not measured.

Aggregation changes the ranking: five supervised models are among the top nine under micro F1 and two under macro F1. GPT-5-mini is second worst among LLMs on micro span F1 (0.450), with precision 0.299, yet its macro score exceeds eight supervised models. Llama-3.3-70B under-extracts: it returns empty lists for 753 of 861 documents. The three `gpt-5.x` models rank below GPT-4.1 under this configuration (GPT-4.1 exceeds GPT-5.2 by +0.058, $p = 0.001$, $p_{\text{Holm}} = 0.086$); this does not isolate reasoning-oriented decoding.

5.2. Performance per Entity Type

On MULTA, the majority class, span F1 is 0.892 for DeepSeek-V4-Flash, 0.855 for GPT-4.1, 0.787 for BERTimbau-large and 0.794 for LegalBert-pt; on RESSARCIMENTO it is 0.793, 0.752, 0.661 and 0.762. On OBRIGACAO the two paradigms tie at the top: DeepSeek-V4-Flash (0.740), BERTimbau-large (0.739) and LegalBert-pt (0.737) are level, with GPT-4.1 at 0.655. The separation is largest on RECOMENDACAO, the rarest class with 22 instances: the leader's group reaches 0.74–0.78, against 0.231 for BERTimbau-large; five supervised models predict no entity of this class, and LegalBert-pt recovers 2 of the 22 (F1 0.160). This class concentrates most of the macro gap and also has the greatest human disagreement (Section 3.4), making sensitivity to the reference annotation especially relevant (Section 5.4).

5.3. Robustness: Matching Criterion, Model Selection, and Human Agreement

Four analyses assess sensitivity to evaluation and selection choices. First, *matching criterion*: IoU thresholds of 0.3 and 0.7 preserve the macro ranking (Spearman 0.988 and 0.916 against the 0.5 ranking), whereas exact matching reverses it (Spearman −0.056). Under exact token matching the encoders lead: LegalBert-pt obtains macro span F1 0.522

Table 3. The central contrast under each reference annotation. Under all three, the best LLM is DeepSeek-V4-Flash and the best supervised model is LegalBert-pt. Observed differences with paired bootstrap 95% CIs; Spearman correlation of the full ranking against the one obtained with A1.

Reference	DeepSeek-V4-Flash	LegalBert-pt	Gap	95% CI	Spearman
A1 (gold)	0.799	0.613	$+0.185$	$[+0.097, +0.261]$	1.00
A2	0.873	0.608	$+0.265$	$[+0.195, +0.331]$	0.96
A3	0.817	0.591	$+0.226$	$[+0.145, +0.303]$	0.98

(micro 0.648) against 0.373 (0.355) for DeepSeek-V4-Flash, and all ten supervised models exceed every LLM on macro F1.

LLM boundary errors are systematic and local. Of the leader's 344 matched spans, only 21 reproduce the gold character offsets exactly (exact micro F1 0.050 in characters against 0.355 in tokens): 195 end one character early (129 omit a final semicolon, 66 a full stop), which the token projection absorbs, and 73 start thirteen characters early by including headings such as *RENOVAÇÃO DA*. Partial matching therefore fits a deployment in which extracted passages are candidates for human review; automatic registry population needs accurate boundaries, where the encoders are clearly better.

Second, *model selection*: rescoring every model on the 688 documents never used for configuration selection leaves the supervised scores essentially unchanged (LegalBert-pt 0.623, BERTimbau-large 0.604), while the leader scores 0.810; the central gap remains $+0.187$ (95% CI $[+0.089, +0.268]$). Third, restricting evaluation to the 202 documents with entities raises the five leading LLMs by 0.03–0.07 macro F1 and the supervised models by at most 0.01, because LLM false positives on empty decisions weigh more; this subset addresses document composition. Fourth, the leader's micro span F1 (0.825) lies within the human pairwise range (0.824–0.838), and its macro score (0.799) exceeds the gold-versus-annotator values (0.792–0.795) but not the agreement between the two independent annotators (0.812). Human agreement is an empirical reference, not a ceiling.

5.4. Sensitivity to the Choice of Reference

The gold is one annotator's judgment, and Section 3.4 shows the other two experts diverge from it systematically on `RECOMENDACAO`, so a conclusion measured against it could be measuring adherence to its annotation policy rather than extraction quality. We test this by rescoring *the same predictions* against each of the three annotations in turn (Table 3).

Rankings remain similar, and the same two models lead their paradigms under all three references. The gap is smallest against the gold and widens against the two annotators who did not produce it, with all three intervals well clear of zero. The widening is concentrated on `RECOMENDACAO`: against A2 and A3, the leader's F1 on this class rises from 0.769 to 0.921 and 0.873, because both annotators record more recommendations addressed to audited entities (33 and 25 spans under the scope guideline) than the gold (22), whereas LegalBert-pt stays between 0.160 and 0.214; the leader also keeps its lead on the three other classes under both annotators. The gold-based gap is thus conservative with respect to the gold's registration threshold for recommendations. This is descriptive robustness on the same corpus, not a test of cross-court generalization.

5.5. Prompting Technique

Chain-of-thought [23] reduces the leader's macro span F1 by 0.097 (0.799 → 0.701), a loss of precision (0.813 → 0.705) rather than recall, and raises GPT-4.1 (0.749 → 0.759), GPT-5.2 (0.692 → 0.750) and Qwen2.5-72B (0.605 → 0.622). A two-stage variant changes the leaders little (DeepSeek-V4-Flash 0.795, GPT-4.1 0.761) and lowers Llama-3.3-70B from 0.250 to 0.140. These point estimates do not isolate model capacity.

5.6. Error Analysis

For the leading model, 344 of the 411 gold entities are recovered with the correct label at IoU ≥ 0.5 (recall 0.837), and 344 of its 423 scored predictions are correct (precision 0.813). Of the 67 missed entities, 37 receive no overlapping prediction and 30 are hit only by predictions that fail the match. Of the 79 incorrect predictions, 37 overlap no gold entity, 38 overlap one below the threshold and 4 match it with the wrong class.

The profile mirrors the human study. Unmatched insertions fall mostly on `OBRIGACAO` (20 of the 37; 6 `MULTA`, 9 `RECOMENDACAO`, 2 `RESSARCIMENTO`), and none is fabricated—every string the leader emits is located in the decision. The nine `RECOMENDACAO` insertions are recommendations that the gold does not register but Annotator 2 registered in all nine cases and Annotator 3 in six: on this class the model sides with the independent annotators against the gold, the same registration-threshold divergence seen in Section 3.4. Class confusion accounts for only 4 of the 146 errors, so presence and boundary errors dominate.

5.7. Limitations

The corpus comes from one court and is concentrated in one year. The gold reflects one annotator's judgment; rescoring against the two other annotations preserves the leading model and widens its margin, and disagreements were not adjudicated. Configuration selection is global rather than nested; rescoring without the development documents shows no optimistic bias, but nested cross-validation was not run. Varying the positive/negative ratio of the prompt is left for future work, as are per-model prompt optimization and the evaluation of RoBERTaLexPT [20] and Sabiá-4 [34]. At Azure list prices, the leader processes the corpus for US$1.94 (US$0.35 with prompt caching), at a median 2.7 s per decision over a 50-document sample, against 0.04 s for LegalBert-pt on an Apple-silicon laptop; reviewer effort and productivity gains were not measured.

6. Conclusion

We introduced `decicontas.br`, a NER dataset of Brazilian state Court of Accounts decisions, with paired pre-/post-review releases, a confident-learning audit, and a three-annotator agreement study. Under IoU ≥ 0.5, DeepSeek-V4-Flash exceeds the strongest supervised baseline, LegalBert-pt, by $+0.185$ macro span F1 after Holm correction. Its lead persists under all three reference annotations, with a wider margin, and its difference from GPT-4.1 is not significant. Exact matching reverses the ranking in favor of the encoders. Few-shot LLMs are the stronger option for proposing candidate passages for human review, whereas automatic registry population would require a boundary fidelity that, in this evaluation, only the encoders approach.

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